Summary of Data Science for Psychologists - 1.4 Vectors

Vectors are the core data structure in R. They are linear sequences of elements of the same type (logical, numeric, character). Each vector has a length, type, and optional attributes.

Inspecting vectors involves functions like length(), typeof(), names(), str(), and is.vector(). Elements can be accessed by position or name.

Vectors are created using c(), :, seq(), and rep(). R coerces mixed types to the most flexible type.

Functions applied to vectors either summarize (sum, mean) or transform (arithmetic, comparisons, logical operations, sorting). Logical values behave numerically (TRUE=1, FALSE=0).

Indexing can be numeric or logical. Numeric indexing selects positions; negative indices exclude. Logical indexing filters based on conditions. which() returns positions of TRUE values. subset() is convenient but drops NAs.

Scalars in R are vectors of length 1.

Practice topics include type coercion, reversing vectors, matching, sorting, generating sequences, filtering, named vectors, and using built-in LETTERS.

Importance: Vectors underpin tibbles, data frames, lists, ggplot inputs, dplyr transformations, and statistical modeling.